4. Aggregate Functions and Grouping Use aggregate functions along with GROUP BY and HAVING clauses to retrieve summarized data from the database.

mysql> CREATE TABLE Student_Result (Student_ID INT,Student_Name VARCHAR(50),Subject VARCHAR(30),Marks INT,Department VARCHAR(30));

Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Student_Result VALUES(1, 'Amit', 'DBMS', 78, 'Computer'),(2, 'Neha', 'DBMS', 85, 'Computer'),(3, 'Rahul', 'OS', 72, 'Computer'),(4, 'Sneha', 'OS', 88, 'IT'),(5, 'Pooja', 'DBMS', 91, 'IT'),(6, 'Ravi', 'OS', 65, 'IT');

Query OK, 6 rows affected (0.00 sec)

Records: 6 Duplicates: 0 Warnings: 0

Aggregate Functions

mysql> SELECT Department, **COUNT**(Student_ID) AS Total_Students FROM Student_Result GROUP BY Department;

```
+------------+----------------+
| Department | Total_Students |
+------------+----------------+
| Computer   |              3 |
| IT         |              3 |
+------------+----------------+
```

2 rows in set (0.00 sec)

mysql> SELECT Subject**, AVG(**Marks) AS Avg_Marks FROM Student_Result GROUP BY Subject;

```
+---------+-----------+
| Subject | Avg_Marks |
+---------+-----------+
```

```
| DBMS    |   84.6667 |
| OS      |   75.0000 |
+---------+-----------+
2 rows in set (0.00 sec)
```

mysql> SELECT Department,**MAX**(Marks) AS Highest_Marks,**MIN**(Marks) AS Lowest_Marks FROM Student_Result GROUP BY Department;

```
+------------+---------------+--------------+
| Department | Highest_Marks | Lowest_Marks |
+------------+---------------+--------------+
| Computer   |            85 |           72 |
| IT         |            91 |           65 |
+------------+---------------+--------------+
2 rows in set (0.00 sec)
```

mysql> SELECT Department, **SUM**(Marks) AS Total_Marks FROM Student_Result GROUP BY Department;

```
+------------+-------------+
| Department | Total_Marks |
+------------+-------------+
| Computer   |         235 |
| IT         |         244 |
+------------+-------------+
2 rows in set (0.00 sec)
```

GROUP BY with HAVING Clause

```
mysql> SELECT Department, AVG(Marks) AS Avg_Marks FROM Student_Result GROUP BY Department HAVING AVG(Marks) > 80;

+------------+-----------+
| Department | Avg_Marks |
+------------+-----------+
| IT         |   81.3333 |
+------------+-----------+
1 row in set (0.00 sec)
```

```
mysql> SELECT Subject, COUNT(Student_ID) AS Student_Count FROM Student_Result GROUP BY Subject HAVING COUNT(Student_ID) > 2;

+---------+---------------+
| Subject | Student_Count |
+---------+---------------+
| DBMS    |             3 |
| OS      |             3 |
+---------+---------------+
2 rows in set (0.00 sec)
```

GROUP BY with WHERE and HAVING

```
mysql> SELECT Department, AVG(Marks) AS Avg_Marks FROM Student_Result WHERE Marks >= 70 GROUP BY Department HAVING AVG(Marks) > 75;

+------------+-----------+
| Department | Avg_Marks |
+------------+-----------+
| Computer   |   78.3333 |
| IT         |   89.5000 |
```

```
+------------+-----------+
2 rows in set (0.01 sec)
```